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BIOS vs. UEFI Comparison

Category	BIOS	UEFI
Definition	BIOS (Basic Input/Output System) is older firmware that initializes the computer's hardware and starts the operating system.	UEFI (Unified Extensible Firmware Interface) is modern firmware that replaces the traditional BIOS and provides more advanced boot and hardware management features.
Boot Process	BIOS performs hardware checks, looks for a bootable device, and loads the bootloader from the disk's boot sector.	UEFI initializes hardware and loads a bootloader from the EFI System Partition. It can also verify boot software using Secure Boot.
Maximum Disk Support	Traditional BIOS commonly works with MBR, which has a practical disk-size limit of about 2 TB for standard 512-byte sectors.	UEFI commonly uses GPT, which supports disks much larger than 2 TB. GPT can support extremely large disks and many partitions.
Partition Style	Mainly uses **MBR (Master Boot Record)**.	Mainly uses **GPT (GUID Partition Table)**.
Security Features	Has limited built-in boot security compared with UEFI. Traditional BIOS does not provide Secure Boot.	Supports security features such as **Secure Boot**, which checks that trusted software is used during the boot process.
Speed	Generally has a slower and more basic boot process.	Generally provides faster boot and resume times.
Advantages	Simple, widely compatible with older hardware and operating systems, and useful for legacy systems.	Supports large drives, more partitions, faster booting, modern security features, and a more flexible firmware environment.
Disadvantages	Limited partitioning and disk support, older technology, and fewer built-in security features.	May have compatibility issues with very old hardware or operating systems. It can also be more complex than traditional BIOS.
Modern Usage	Mostly found on older computers or used through legacy/compatibility modes.	Used on most modern computers and is the standard firmware environment for current systems.

Why UEFI Has Largely Replaced BIOS

BIOS and UEFI both perform an important job when a computer starts. They initialize the computer's hardware and then begin the process of loading the operating system. However, BIOS is an older technology that was designed for computers with much simpler hardware and storage requirements. UEFI was developed as a replacement for the traditional BIOS and provides features that are better suited to modern computers. Microsoft describes UEFI as the replacement for the older BIOS firmware interface.

One major difference between BIOS and UEFI is how they handle disk partitions. Traditional BIOS normally works with the MBR partition style. MBR has limitations on disk size and the number of primary partitions. UEFI normally works with GPT, which supports much larger drives and many more partitions. The UEFI specification also explains that GPT provides advantages such as support for many partitions, backup partition tables, and improved data integrity through CRC32 checks.

UEFI also provides better security. One of its most important features is Secure Boot. Secure Boot checks the digital signatures of boot software and allows the computer to start only trusted software. This helps protect the computer from certain types of bootkits and other malware that try to load before the operating system.

Another advantage of UEFI is improved boot performance. Modern UEFI systems can provide faster boot and resume times than older BIOS-based systems. UEFI also provides a more flexible environment for manufacturers to add features and tools. Modern UEFI interfaces can include graphical menus and support mouse navigation, although the exact features depend on the computer manufacturer.

BIOS is still useful for older computers and legacy operating systems because of its compatibility. Some modern computers also provide a compatibility mode that allows older BIOS-style booting. However, new operating systems and modern computer hardware are designed mainly around UEFI and GPT. For example, Microsoft requires UEFI and GPT for Windows installations using the modern UEFI boot mode.

Overall, UEFI has largely replaced BIOS because it is better suited to modern computing. It supports larger storage devices, more partitions, faster booting, and stronger security features such as Secure Boot. BIOS remains important for understanding older computer systems, but UEFI is the preferred firmware technology for modern computers because it provides greater flexibility, compatibility with newer hardware, and improved security.

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